

Proposal for High-Performance Computing (HPC) Workstation for High-Energy Astrophysics and X-ray Astronomy Research

Estimated Cost: INR 10,00,000

Introduction:

High-energy astrophysics and X-ray astronomy are rapidly advancing fields that require significant computational power to analyze large datasets, simulate complex physical processes, and perform data-intensive simulations. To facilitate cutting-edge research in these areas, we propose the acquisition of a high-performance computing workstation.

Purpose:

The high-performance computing workstation will serve as a dedicated resource for researchers and scientists engaged in high-energy astrophysics and X-ray astronomy studies. This system will enable faster data processing, complex simulations, and efficient modeling, ultimately advancing our understanding of the universe's most energetic and elusive phenomena.

Components and Specifications:

1. **CPU:** Dual Intel Xeon processors (32 cores) for parallel processing capabilities - INR 3,00,000.
2. **GPU:** NVIDIA GeForce RTX A6000 with 48 GB VRAM for accelerated scientific computing - INR 2,50,000.
3. **RAM:** 256 GB DDR4 ECC RAM for handling large datasets and simulations - INR 1,50,000
4. **Storage:** 2 TB NVMe SSD for fast data access and 8 TB HDD for storage - INR 1,00,000.
5. **Motherboard:** High-end motherboard with support for multiple GPUs and high-speed connectivity - INR 50,000.
6. **Cooling:** Efficient cooling solution to ensure stable operation - INR 20,000.
7. **Power Supply:** High-wattage PSU for reliable performance - INR 15,000.
8. **Monitor:** 4K Ultra HD monitor for data visualization and analysis - INR 40,000.
9. **Peripherals:** Keyboard, mouse, and other necessary accessories - INR 10,000.
10. **Operating System and Software:** Licensing for scientific software packages and operating system - INR 30,000.
11. **Warranty and Support:** Extended warranty and support for three years - INR 25,000.
12. **Installation and Setup:** Professional setup and installation - INR 10,000.

Total Estimated Cost: **INR 10,00,000**

Justification:

Enhanced Computational Power: The proposed workstation offers exceptional computational power with dual Xeon processors and a high-end GPU, allowing researchers to run resource-intensive simulations and data analysis tasks efficiently.

Large RAM and Storage: The substantial RAM and storage capacity are crucial for handling large datasets commonly encountered in high-energy astrophysics and X-ray astronomy research.

Accelerated Processing: The GPU is specifically chosen for its capabilities in accelerating scientific computing, making it ideal for simulations and data processing in these fields.

Reliability and Warranty: The extended warranty and professional support ensure the system's reliability, reducing downtime and potential data loss.

High-Resolution Visualization: The 4K Ultra HD monitor facilitates detailed data visualization, enabling researchers to gain deeper insights into their findings.

Compatibility: The workstation will be configured to work seamlessly with specialized astrophysics and astronomy software, ensuring compatibility and optimal performance.

Though, the HPC workstation has been proposed for mainly research in Astronomy and Astrophysics, it can also be used for research in the field of Materials Science, Nanotechnology, and AI and Machine Learning etc.

Investing in this high-performance computing workstation will empower our researchers to push the boundaries of knowledge in high-energy astrophysics and X-ray astronomy. It will lead to breakthroughs in our understanding of the universe's most energetic and mysterious phenomena, making it a sound investment for our institution.